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Modulating SEI formation *via* tuning the solvation sheath for lithium metal batteries†

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The solvation sheath of Li^+–glyme was modulated to enhance Li^+–$TFSI^-$ association by adopting a highly polar solvent, especially water molecules, which affects the solid electrolyte interface (SEI) layer composition. By the Li^+–$TFSI^-$ association, a $TFSI^-$ anion-derived SEI layer is formed on the Li metal anode, resulting in higher Li metal anode efficiency.

Despite the significant advantage of Li metal anodes for Li batteries, unstable solid electrolyte interphase (SEI) layer formation and low Coulombic efficiency (C.E.) caused by the uncontrollable dendritic growth have limited their practical use for Li–metal batteries (LMBs).[1,2] To address these issues, significant research has been reported including electrolyte engineering to stabilize the SEI layer and the introduction of an artificial SEI layer or Li protection layer to suppress side reactions between the electrolyte and Li metal anode.[3–6] In particular, engineering of the solvation sheath structure of Li-ions in the electrolyte has been vastly explored due to its simplicity and effectiveness to manipulate the composition, structure, and morphology of the SEI layer.[7–10] In general, Li ions are directly bound to oxygen atoms in solvents with 4–6 coordinate sites, while anions are free in the electrolyte without joining the solvation sheath.[11] In this circumstance, migration of charged species of solvated Li^+ and free-anions by applied potential causes anion depletion at the Li side.[12] Therefore, the SEI layer is composed of reduced organic components, mainly. The reduced organic component-based SEI layer induces localized current, dendritic Li deposition, and breakdown of the SEI layer, resulting in cell failure during repeated cycling.[13]

Herein, we report that a water molecule induces anion-derived solid electrolyte interface (SEI) layer formation, in ether-based electrolytes composed of diethylene glycol dimethyl ether (DEGDME) and lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) salt. The DEGDME is selected considering its structure, which has three ether groups in a linear chain. Therefore, all ether groups in the two DEGDMEs bind to all coordination sites of Li^+, minimizing other structural effects. The LiTFSI has no effect on solvents and water due to its chemical stability. In the electrolyte composition, we confirmed that the water molecules effectively disturb the $(DEGDME)_2(Li^+)$ solvation sheath. Therefore, $TFSI^-$ anions interact with the Li^+ in the disrupted sheath, resulting in Li^+–$TFSI^-$ association. The Li^+–$TFSI^-$ association behavior in the electrolyte is characterized by Raman spectroscopy and a viscometer. The SEI layer formation behavior and its chemical components are confirmed by linear sweep voltammetry (LSV) and X-ray photoelectron spectroscopy (XPS), which reveals that a LiF-rich SEI layer was formed. Resulting from the LiF-rich SEI layer, uniform and dendrite-free Li plating characteristics are monitored by scanning electron microscopy (SEM). The electrochemical properties of the Li anode with the LiF-rich SEI layer are evaluated using a Li symmetric cell and a Li (20 μm)‖$LiFePO_4$ (LFP) full cell. These results confirmed that water affects SEI layer formation on the Li metal surface by controlling the solvation sheath structure and degree of Li^+–$TFSI^-$ association of the electrolyte.

Scheme 1 describes the molecular dynamics (MD) simulation snapshots of the solvation sheath structure *via* anhydrous and water-containing electrolytes (Scheme 1a) and a schematic illustration of the SEI layer formation and Li deposition behaviours by the solvation sheath structures (Scheme 1b). As shown in Scheme 1a, the solvation sheath structure and Li^+–$TFSI^-$ association differ according to the amount of added water molecules. In the case of an anhydrous electrolyte, a stable solvation sheath is formed by two DEGDME molecules and one centred Li^+, in which six coordination sites of the Li^+ are bound to the oxygen atoms in DEGDME molecules. The DEGDME molecules inhibit further association or coordination with other molecules, especially $TFSI^-$ anions (shielding effect).[14]

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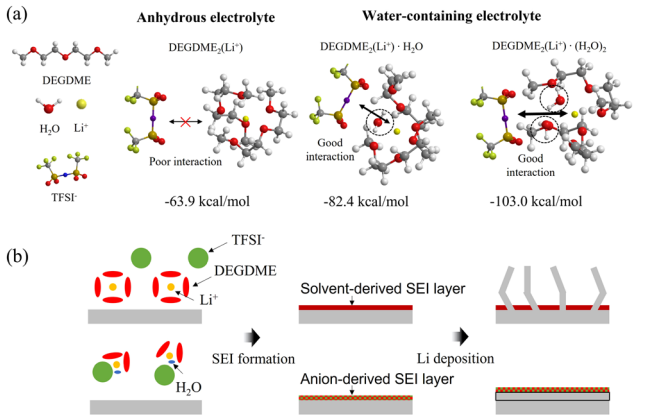

Scheme 1 Concept of the engineering Li^+–glyme solvation structure by water molecules. (a) Snap-shot images during molecular dynamics simulation. (b) Schematic illustration of SEI layer formation by the solvation sheath tuning.

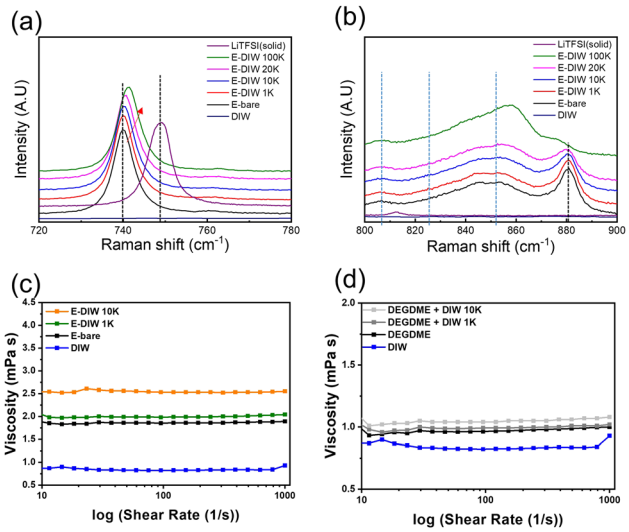

Fig. 1 Raman spectrum (a and b) and viscoelastic behaviour (c and d) analysis to monitor the degree of Li^+–$TFSI^-$ association according to the amount of added DIW.

On the other hand, in the case of water-containing electrolyte, the solvation sheath becomes loosened and the solvated Li^+ readily attracts the $TFSI^-$ electrostatically, by a water molecule (Li^+–$TFSI^-$ association).[15] As a result of the association, the associated $TFSI^-$ ions can be reduced during the SEI formation, even with an applied potential. In an anhydrous electrolyte, the applied potential repels $TFSI^-$ anions from the Li metal to the cathode, resulting in an organic-rich SEI layer by the reduction of solvent bound to Li^+ (solvent-derived SEI layer). The organic-rich SEI layer has poor mechanical properties and Li^+ flux, resulting in localized Li^+ deposition and Li dendrites.[16] Meanwhile, in the case of water-containing electrolytes, the Li^+–$TFSI^-$ association maintains a high concentration of $TFSI^-$ ions in the vicinity of the Li metal surface, even with an applied potential, resulting in an inorganic-rich SEI layer (anion-derived SEI layer) and dendrite-free Li deposition (Scheme 1b).

To identify the degree of Li^+–$TFSI^-$ association, Raman spectroscopy analysis was conducted according to the deionized water (DIW) addition. Fig. 1(a and b) represents the Raman spectra of LiTFSI, DIW, and electrolytes as a function of ppm of additional DIW (hereafter, the bare electrolyte, and 1000, 10 000, 20 000, and 100 000 ppm DIW added electrolytes were denoted E-bare, E-DIW 1k, E-DIW 10k, E-DIW 20k and E-DIW 100k, respectively). Li^+–$TFSI^-$ association was evaluated by monitoring the S–N–S bending vibration peak at the region of 700–800 cm^{-1} (Fig. 1(a)). The Raman shifts of the fully dissociated $TFSI^-$ (bare electrolyte) and that associated with Li^+ (solid LiTFSI) were assigned at 739.9 cm^{-1} and 749.2 cm^{-1}, respectively. Interestingly, the S–N–S bending vibration peaks shift gradually with a redshift from 739.9 cm^{-1} to 741.5 cm^{-1} as the DIW content was increased. As shown in Fig. 1(b), the Raman peaks of the free-DEGDME (un-solvated with Li^+) were increased at 805, 825, and 850 cm^{-1}, while those of the solvated DEGDME were decreased according to the addition amount of DIW at 880 cm^{-1}.[17] The water molecules exist in the form of isolated monomers to pentamers rather than bulk water (Fig. S1, ESI†). All the Raman spectroscopy results suggest that water molecules enhance the Li^+–$TFSI^-$ association by disrupting the Li^+–glyme solvation sheath.

The viscometer analysis of the electrolytes was conducted to analyse their viscoelastic response according to the addition of DIW (Fig. 1(c)). Interestingly, as the DIW was added to the ether-based electrolyte, the viscoelasticity showed an increasing tendency, which was opposite to previous results with ionic liquid-based electrolytes.[18] Even with the lower viscosity value of the bulk DIW than the ether electrolytes in the overall shear rate, the viscosities of E-DIW 1k (2.00 mPa S) and E-DIW 10k (2.54 mPa S) were higher than that of the bare electrolyte (E-bare, 1.86 mPa S). The increasing viscosity implied that the attractive force between the Li^+ cations and $TFSI^-$ anions was generated by the disruption of the normal solvation sheath *via* the addition of DIW. To compare with the DEGDME–DIW interaction, the viscoelastic behaviors of neat DEGDME and DIW were investigated (Fig. 1(d)). The viscosity of DEGDME showed almost no differences when DIW was added. The results indicate that the viscosity change with the addition of DIW was influenced more by the interaction between the salts than by that of the solvents.

To evaluate the dependence of Li nucleation and deposition behaviour on DIW concentration, scanning electron microscopy (SEM) analysis was conducted (Fig. 2). In the E-bare electrolyte, relatively large Li nuclei and dendrites formed (Fig. 2(a–c)), which grew in a small number of areas resulting from several initial nucleation sites. Even with the low specific capacity, the dendrites grew to ~8 μm long (Fig. 2(c)). In contrast, Li deposition in E-DIW 1k and E-DIW 10k showed an increased number of nucleation sites with smoother particle morphology (Fig. 2(d–f) and (g–i) correspond to E-DIW 1k and E-DIW 10k, respectively). The smooth and homogeneous Li deposition is attributed to the homogeneous Li^+ flux, which resulted from the inorganic rich anion-derived SEI layer.[19]

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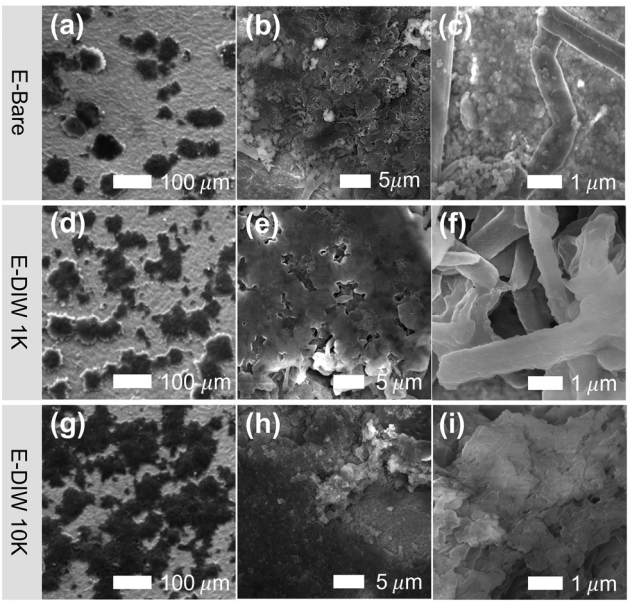

Fig. 2 Li deposition behavior on Cu foil with a capacity of 0.1 mA h cm^{-2} was observed using SEM.

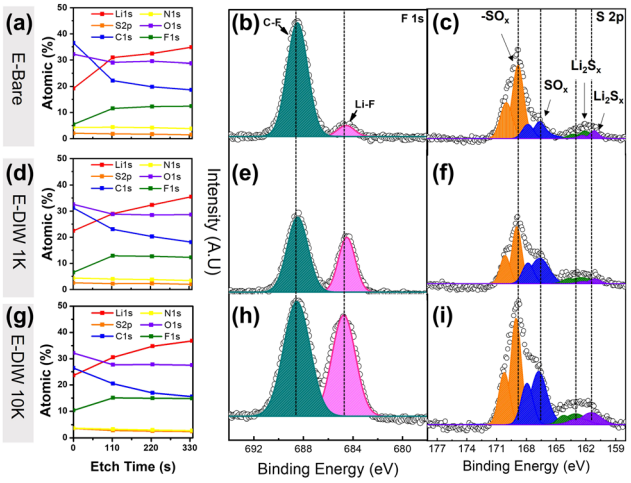

Fig. 4 Characterization of the compositions of the surface layer on the Li metal anodes in the Li||LFP coin cells after 1 cycle at 1C including formation cycles using E-bare (a–c), E-DIW 1k (d–f), and E-DIW 10k (g–i). (a, d and g) Atomic percentage concentrations of different elements with increasing etching time. (b, e and h) Corresponding F 1s (c, f and i) and S 2p XPS spectra.

The LSV analysis of the ether-based electrolytes is shown in Fig. S2 (ESI†). The peak at 1.85 V seen on the LSV curves was caused by the reduction of the $TFSI^-$ anions in the electrolytes.[20] E-bare showed a specific peak in that potential area, but as the DIW content increased, the peak corresponding to $TFSI^-$ reduction was widened. Based on the previous reports,[21] in which the $TFSI^-$ reduction potential is shifted to a lower value and broadened when the Li^+–$TFSI^-$ association is increased, the LSV behaviour is in good accordance with the previous report. In addition, the DEGDME reduction peaks at 0.52 and 0.84 V disappeared in the E-DIW 10k results, which suggested that the $TFSI^-$ anions were preferentially reduced.[22] This result was confirmed with the following XPS results (Fig. 4).

To verify the effect of DIW on the electrochemical performance, a cycling test was performed with a Li symmetrical cell and an LFP full cell (Fig. 3). Fig. 3(a) shows the Li symmetric cycling performance. The initial overpotentials of E-bare, E-DIW 1k, and E-DIW 10k were 36, 46, and 43 mV, respectively. After repeated plating and stripping, the overall cell polarization increased with stable cycling performance even increasing the amount of DIW (Fig. 3(b)). The difference was apparent in the LFP full cell cycling performance (Fig. 3(c)). The LFP full cells were investigated by using extremely thin Li metal (20 μm, 4 mA h cm^{-2}) and a 2 mA h cm^{-2} LFP cathode (N/P = 2). The evaluation using the thin Li had the advantage that it could be interpreted as maximizing the performance of the Li metal anode rather than the performance of the cathode. As shown in Fig. 3(c), E-bare showed rapid degradation of capacity over ~20 cycles. However, E-DIW 1k performed stably beyond 20 cycles and started to degrade at ~32 cycles. In E-DIW 10k, the cell ran well until 67 cycles and the capacity faded relatively slowly. As shown in Fig. 3(d), E-DIW 10k had better efficiency than the others, and up to 95.3% Coulombic efficiency was achieved in the 43rd cycle. Notably, there was an obvious capacity degradation for E-DIW 10k in the early cycles compared to E-bare and E-DIW 1k. The continuous decrease in capacity in the initial cycles is thought to be due to the degradation of the cathode materials by water molecules (Fig. S3, ESI†).

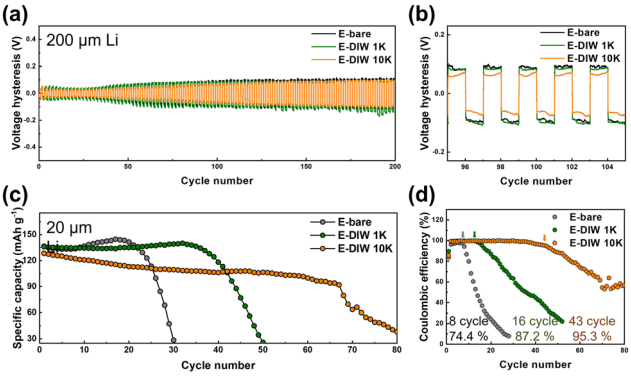

Fig. 3 The cell performance tests using ether-based electrolytes were carried out with a Li symmetrical cell (a and b) and an LFP full cell (c and d).

The LFP full cell cycle test using carbonate-based electrolytes containing $LiPF_6$ salt was also evaluated (Fig. S4 (ESI†), bare carbonate electrolyte, 1000, and 10 000 ppm DIW added electrolyte were denoted C-bare, C-DIW 1k, and C-DIW 10k, respectively). In the case of carbonate electrolyte, there is rapid capacity fading. Even worse was C-DIW 10k, which could not start cycling. The hydrolysis reaction between water and $LiPF_6$ degraded the water-containing carbonate electrolyte (Fig. S5, ESI†). This suggested that the water in the carbonate electrolyte with $LiPF_6$ could not be applied to solvation sheath engineering.

XPS was performed to examine the anion-derived LiF-rich SEI layer by the water-containing electrolytes on the LFP

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cathode and the Li anode surface after 1 cycle at 1C of the formation process (1 cycle at 0.1C and 0.5C, respectively). First, as shown in Fig. S6 (ESI†), the XPS spectra of the LFP cathode surface revealed that the more the DIW content increased, the more prominent the peak at 724 eV appeared. According to previous reports, the Fe 2p spectra of LFP after storage show peaks at 710 eV for Fe $2p_{3/2}$ and 725 eV for Fe $2p_{1/2}$ as a result of spin-orbit coupling.[23] The peak at 724 eV was not present in the E-bare spectra but appeared in the Fe 2p spectra of E-DIW 1k and increased in size for E-DIW 10k, which indicates that the degradation of LFP increased as the content of added water increased. As shown in Fig. 4(a), (d), and (g), the surface atomic percentage profiles over the etching time changed depending on the DIW content. As the DIW content increased, the F 1s and Li 1s atomic percentage showed an increasing tendency overall on the Li anode surface layer. In contrast, the C atom percentage tended to be significantly lower, and the S atomic percentage showed a slight lowering tendency. The change of the SEI layer after DIW addition was shown by the F 1s XPS spectra (Fig. 4(b), (e) and (h)). In Fig. 4(b), E-bare showed a considerably low intensity at 684.6 eV, corresponding to the Li–F bond energy. However, as the amount of DIW in the electrolyte increased, the Li–F intensity increased to a point comparable to the C–F bond peak.[24] Also, the S 2p XPS spectra in Fig. 4(c), (f) and (i) showed an overall increase in S-containing components with DIW content, which could have been the cause of the $TFSI^-$ reduction products. Based on the XPS results, we can infer that the $TFSI^-$ anion binding with the Li solvation structure was transferred to the Li anode side together and reduced into small species like LiF, Li_2SO_4, $Li_2S_2O_4$, and Li_2S, during the reduction process.[25] In the anion-derived SEI layer, various reduction products formed on the Li surfaces, which managed not only to maintain the Li cycling performance but also to improve it. This anion-derived passivation layer seemed to exhibit similar characteristics to the concentrated electrolyte in an ion transfer mechanism. However, if water additive was applied in the LMBs, the development of a cathode material having moisture stability should be achieved, as shown in Fig. S3 (ESI†).

In summary, our research has confirmed the effects of water-containing electrolytes on LMBs in terms of the solvation sheath of the Li^+ and SEI layer composition. Resulting from the high polarity of the water molecules, the water molecules disrupt the Li^+–DEGDME solvation sheath formation, effectively. This disruption changes the solvation structure, providing open Li^+ coordination sites and Li^+–$TFSI^-$ association. During charging the cell, the SEI layer is formed with associated $TFSI^-$ anion**s**, resulting in the LiF-rich SEI layer. The Li anode with the $TFSI^-$-derived LiF-rich SEI layer brings reduced surface resistance and long-term cycling stability compared to an anhydrous electrolyte. When solving the oxidation problem of the cathode materials, the water can act as an interesting additive in the LiTFSI electrolyte system. All the above results demonstrate that it is a promising research subject for solvation structure engineering in battery electrolytes and stable LMBs.

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Conflicts of interest

There are no conflicts to declare.

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